Matthew Kwan | CV

Citizenship	Australian
Date of Birth	8 January 1992
Email	matthew.a.kwan@gmail.com
Address	Am Campus 1, A-3400 Klosterneuburg, Austria

Employment & Education

Professor
Institute for Science and Technology Austria (ISTA)
2021 – present (awarded tenure in 2026)

Szegő Assistant Professor
Stanford University
2018 – 2021

Doctor of Science in Mathematics
With ETH medal
ETH Zürich
2014 – 2018
Adviser: Benny Sudakov

Bachelor of Commerce and Bachelor of Science (Advanced Mathematics)
With first-class honours and the university medal in pure mathematics
University of New South Wales
2010 – 2014

Awards & Scholarships

Recent prizes

- Frontiers of Science Award, category "Combinatorics", 2026
- Frontiers of Science Award, category "Discrete Geometry and Graph Theory", 2026
- Prize of the Austrian Mathematical Society (ÖMG Förderungspreis), 2024
- SIAM Dénes Kőnig Prize, 2020

Major grants and scholarships

- European Research Council (ERC) Starting Grant "RANDSTRUCT", 2023 – 2028
  (project number 101076777; EUR €1 343 890)
- USA National Science Foundation (NSF) standard grant, 2020 – 2023
  (award number 1953990; US $179 217)
- Swiss National Science Foundation (SNSF) Early Postdoc.Mobility Fellowship, 2018 – 2020
  (project number 178493; US $75 150)

Teaching

ISTA

- Introduction to the Probabilistic Method (2022)
- Extremal Graph theory (2023)
- Ramsey Theory (2024)
- Additive Combinatorics (2025)
- Mathematics Track Core Course (2026, 2027)
- AI for Mathematics (2027)

Stanford University

- Introduction to Probability (2019, 2021)
- Graph Theory (2019)
- Discrete Probabilistic Methods (2019, 2020, 2021)
- Introduction to Combinatorics and its Applications (2021)

Special Programs

- Anti-concentration and combinatorics
  (Czech Summer School on Discrete Mathematics, 2026)

Academic Service

PhD students supervised

- Yiting Wang (2023 – present)
- Farhood Rostamkhani (2023 – present)
- Alex Grebennikov (2025 – present)
- Lenka Kopfová (2025 – present)

Postdocs mentored

- Walner Mendonça (2021 – 2022)
- Michael Anastos (2022 – 2026)
- Oliver Cooley (2022 – 2023)
- Benjamin Moore (2023 – 2025)
- Lyuben Lichev (2024 – 2025)
- Kalina Petrova (2024 – present)
- Zhihan Jin (2026 – present)

Event organisation

- Co-organiser of a 2028 *Junior Trimester Program* on extremal and probabilistic combinatorics, at the Hausdorff Research Institute for Mathematics in Bonn
- Co-organiser of the 14th *Austrian Stochastics Days*, in 2026 at ISTA
- Co-organiser of the 22nd *Random Structures and Algorithms* conference, in 2026 in Vienna

- Co-organiser of the *Vienna Combinatorics Seminar* (a cooperation between ISTA, TU Vienna and the University of Vienna), 2026 – present
- Co-organiser of the *Combinatorial Tripoint Meeting* (a cooperation between ISTA, the University of Vienna, TU Graz, Masaryk University, the University of Passau and Leipzig University), 2023 – present
- Co-organiser of a 2024 workshop "Probability meets Combinatorics" at ISTA

Internal seminar organisation

- Co-organiser for the ISTA Discrete Analysis Days, 2025 – present
- Co-organiser for the ISTA Combinatorics, Geometry and Topology Seminar, 2021 – present

Institutional responsibilities

- Co-organiser for the ISTA Mathematics Colloquium, 2024 – 2025
- External member of the ISTA faculty recruitment committee for "information and system sciences", 2023
- Member of the ISTA Women Search Committee, 2022 – 2024
- Member of the ISTA Interdisciplinary Projects Committee, 2022
- Member of the ISTA Institute Colloquium Committee, 2021, 2024 and 2026

Other professional activities

- Since 2026, I am an editor for *Advances in Combinatorics*.
- I was on the nomination committee of the *Frontiers of Science* awards for the 2026 International Congress of Basic Science (ICBS).
- I have served as a referee for many journals, conferences, grant agencies and hiring committees.

Outreach

- I participated in interviews with the science writers at ISTA and with *Quanta Magazine*, regarding my resolution of Erdős' conjecture on high-girth Steiner triple systems. This resulted in the publication of two general-audience articles on my research.
- I participated in interviews for the *Geistesblitz* column in *der Standard*, and the *Junge Forschung* column in *die Presse* (these are two Austrian newspapers) about my general research directions.
- I wrote a general-audience expository article about the polynomial Littlewood–Offord problem, for the newsletter of the Austrian Mathematical Society (*Internationale Mathematische Nachrichten*).
- I gave introductory lectures about Ramsey theory and the probabilistic method for high school students on several different occasions (at a 2022 training camp for the Austrian, Czech, Slovak, Polish and Ukrainian teams for the International Mathematics Olympiad, and at summer schools for Ukrainian high school students in 2022 and 2023).

Talks

Invited talks at conferences/workshops

- Discrete Geometry and Graph Theory section, International Congress of Basic Science (Beijing Institute for Mathematical Sciences and Applications). August 2026.
- Combinatorics section, International Congress of Basic Science (Beijing Institute for Mathematical Sciences and Applications). August 2026.
- Innovations and Challenges in Extremal Combinatorics, Institute for Mathematical Sciences (National University of Singapore). August 2026.
- Workshop on Combinatorics, Mathematisches Forschungsinstitut Oberwolfach. January 2026.
- Workshop on Combinatorics, Probability and Computing, Mathematisches Forschungsinstitut Oberwolfach. September 2025.
- Annual meeting of the Austrian Mathematical Society (ÖMG) and the German Mathematical Society (DMV), Johannes Kepler University Linz. September 2025.
- Research program on Extremal and Probabilistic Combinatorics, IAS / Park City Mathematics Institute. July 2025.
- Discrete, Interactive and Algorithmic Mathematics, Algebra and Number Theory, 60th Netherlands Mathematical Congress. April 2025.
- Introductory Workshop on Graph Theory, Simons Laufer Mathematical Sciences Institute (SLMath / MSRI). February 2025.
- New Frontiers in Extremal and Probabilistic Combinatorics, SwissMAP Research station. January 2025.
- Workshop on developments in Combinatorics, Institute for Basic Science. November 2024.
- New Frontiers in Probabilistic and Extremal Combinatorics (workshop of the Clay Research Conference), Clay Mathematics Institute (Oxford University). October 2024.
- Synergies of Combinatorics and theoretical computer science, Bernoulli Center (EPFL). August 2024.
- Minisymposium on Extremal Combinatorics, British Combinatorial Conference. July 2024.
- Discrete Probability Days, Centre de Recerca Matemàtica. October 2023.
- Workshop on random graphs, Technische Universität Dortmund. September 2023.
- European Conference on Combinatorics, Graph Theory and Applications (EUROCOMB 2023) (plenary speaker), Charles University. August 2023.
- Workshop on Random Graphs, Mathematisches Forschungsinstitut Oberwolfach. March 2023.
- Workshop on Combinatorics, Mathematisches Forschungsinstitut Oberwolfach. January 2023.
- Symposium on Discrete Mathematics (for the discrete mathematics group of the German Mathematical Society), Technische Universität Hamburg. September 2022.
- Recent advances in probabilistic and extremal combinatorics, Congressi Stefano Franscini (ETH Zürich). August 2022.
- Third Southwestern German Workshop on Graph Theory (plenary speaker), Heidelberg University. June 2022.
- Colloquia in Combinatorics, London School of Economics. May 2022.
- Workshop on Combinatorics, Probability and Computing, Mathematisches Forschungsinstitut Oberwolfach. April 2022.
- Interfaces of the Theory of Combinatorial Limits, Erdős Center (Rényi Institute). March 2022.
- Dénes König Prize Lecture, SIAM Conference on Discrete Mathematics (DM21). July 2021.

- Special Session on Structural and Extremal Graph Theory, AMS Fall Southeastern Sectional Meeting. October 2020.
- Probabilistic Combinatorics Online 2020 (plenary speaker). September 2020.
- Workshop on Combinatorics, Mathematisches Forschungsinstitut Oberwolfach. January 2020.
- Workshop on Probabilistic and Extremal Combinatorics, Banff International Research Station for Mathematical Innovation and Discovery (BIRS). September 2019.
- BennyFest (celebrating Benny Sudakov's 50th birthday), ETH Zürich. July 2019.
- Workshop on Combinatorics, Probability and Computing, Mathematisches Forschungsinstitut Oberwolfach. April 2019.
- Minisymposium on Analytic and Probabilistic Techniques in Combinatorics, SIAM Conference on Discrete Mathematics (DM18), University of Colorado, Denver. June 2018.

Departmental seminars/colloquia

- Combinatorics Seminar, Cambridge University. October 2026.
- Mathematics Colloquium, Yonsei University. December 2025.
- Discrete Mathematics Seminar, Institute for Basic Science. December 2025.
- Combinatorics Seminar, California Institute of Technology. October 2025.
- Research Seminar in Discrete Mathematics, Beijing Institute for Mathematical Sciences and Applications. April 2025.
- Hausdorff Colloquium, Hausdorff Center for Mathematics (University of Bonn). December 2024.
- Mathematical Physics and Analysis Seminar, ISTA. June 2024.
- Research Seminar in Discrete Mathematics, Beijing Institute for Mathematical Sciences and Applications. February 2024.
- Combinatorics and Discrete Probability Seminar, University of Illinois Chicago. February 2024.
- Combinatorics Seminar, Stanford University. February 2024.
- Browning group (Number Theory) Working Seminar, ISTA. January 2024.
- Spectral Geometry in the Clouds (seminar on spectral theory and geometric analysis). December 2023.
- Inn'formal Probability Seminar, University of Innsbruck. November 2023.
- Informatics and Mathematics Colloquium, University of Passau. July 2023.
- (Not So) Informal Probability Seminar, University of Vienna. April 2023.
- Groningen Probability Seminar, University of Groningen. February 2023.
- MIT–Harvard–MSR Combinatorics Seminar, Massachusetts Institute of Technology. January 2023.
- Combinatorics, Geometry and Topology Seminar, ISTA. June 2022.
- (Not So) Informal Probability Seminar, University of Vienna. May 2022.
- Discrete Mathematics Seminar, Princeton University. April 2022.
- Graz Combinatorics Seminar, Technische Universität Graz. March 2022.
- Combinatorics Seminar, University of Birmingham. February 2022.
- Discrete Seminar, Umeå University. February 2022.
- Graz Combinatorics Seminar, Technische Universität Graz. December 2021.
- Informatics Colloquium, Masaryk University. November 2021.
- Research Seminar in Combinatorics, Tel Aviv University. November 2021.

- Big Seminar, Laboratory of Combinatorial and Geometric Structures, Moscow Institute of Physics and Technology. November 2021.
- Vienna Probability Seminar. November 2021.
- Discrete Mathematics Seminar, Technische Universität Wien. October 2021.
- CMSA (Combinatorial Mathematics Society of Australasia) Seminar. September 2021.
- Copenhagen–Jerusalem Combinatorics Seminar. July 2021.
- Combinatorics and Probability Seminar, Ohio State University. March 2021.
- MIT–Stanford Combinatorics Reading Seminar. March 2021.
- Mathematics Department Colloquium, University of California, Berkeley. February 2021.
- Webinar on Applied Analysis, Max Planck institute for Mathematics in the Sciences. November 2020.
- Graph Theory Seminar, Georgia Institute of Technology. October 2020.
- Graph Theory and Combinatorics Seminar, University of Illinois. September 2020.
- MIT–Stanford Combinatorics Reading Seminar. July 2020.
- Extremal and Probabilistic Combinatorics webinar. July 2020.
- Matroid Union Seminar. June 2020.
- Probability Seminar, Stanford University. November 2019.
- Combinatorics Seminar, Massachusetts Institute of Technology. February 2019.
- Combinatorics Seminar, Emory University. November 2018.
- Mathematics Department Colloquium, Stanford University. November 2018.
- Combinatorics Reading Seminar, Stanford University. October 2018.
- Combinatorics Seminar, Stanford University. October 2018.
- Mittagsseminar (Theory of Combinatorial Algorithms), ETH Zürich. March 2018.
- Horowitz Seminar on Probability, Ergodic Theory and Dynamical Systems, Tel Aviv University. June 2017.
- Graduate Seminar in Probability, ETH Zürich. May 2017.
- Mittagsseminar (Theory of Combinatorial Algorithms), ETH Zürich. November 2016.
- Mittagsseminar (Theory of Combinatorial Algorithms), ETH Zürich. April 2016.
- Pure Mathematics Seminar, University of New South Wales. August 2015.
- Discrete Mathematics Seminar, Monash University. August 2015.
- Algebra and Topology Seminar, Australian National University. August 2015.
- Mittagsseminar (Theory of Combinatorial Algorithms), ETH Zürich. March 2015.

Contributed conference talks

- 19th International Conference on Random structures and Algorithms, ETH Zürich. July 2019.
- European Conference on Combinatorics, Graph Theory and Applications (EUROCOMB 2017), TU Wien. August 2017.
- 18th International Conference on Random structures and Algorithms, Gniezno. August 2017.
- SIAM Conference on Discrete Mathematics (DM16), Georgia State University. June 2016.
- European Conference on Combinatorics, Graph Theory and Applications (EUROCOMB 2015), University of Bergen. August 2015.
- 17th International Conference on Random structures and Algorithms, Carnegie Mellon University. July 2015.
- 37th Australasian Conference on Combinatorial Mathematics and Combinatorial Computing, University of Western Australia. December 2013.

Publications

62. C Greenhill, L Kopfová, M Kwan and K Petrova. On the probability that a random Latin square is all-even. Submitted.
61. M Kwan and H T Pham. Nonlinear lower-tail large deviations at criticality. Submitted.
60. A Ghosal, R Goenka, A Grebennikov, P Keevash, M Kwan and H T Pham. No-$(k+1)$-in-line problem for $k \geq 3$. Submitted.
59. Z Hunter, M Kwan and L Sauermann. Permanents of random matrices over finite fields. Submitted.
58. A Grebennikov and M Kwan. Edge-statistics beyond $1/e$. Submitted.
57. Z Hunter, M Kwan and L Sauermann. Exponential anticoncentration for the permanent. Submitted.
56. M Kwan, K Petrova and M Sawhney. Parities in random Latin squares. Submitted.
55. A Grebennikov and M Kwan. Geometric Littlewood–Offord problems via lattice point counting. Submitted.
54. Z Jin, M Kwan, L Sauermann and Y Wang. Algebraic aspects of the polynomial Littlewood–Offord problem. Submitted.
53. Z Jin, M Kwan and L Lichev. Colouring random Hasse diagrams and box-Delaunay graphs. Submitted.
52. M Anastos, M Kwan and B Moore. Smoothed analysis for graph isomorphism. Submitted. A conference version appeared at *STOC 2025*.
51. M Glasgow, M Kwan, A Sah and M Sawhney. The exact rank of sparse random graphs. *Journal of the European Mathematics Society*, to appear.
50. A Grebennikov and M Kwan. No-$(k+1)$-in-line problem for large constant k. *Bulletin of the London Mathematical Society* 58.10 (2026).
49. V Jain, M Kwan and M Michelen. Entangled states are typically incomparable. *Communications in Mathematical Physics* 407.214 (2026).
48. Z Hunter, M Kwan, L Sauermann and M Sawhney. On random matrices with large corank. *International Mathematics Research Notices* 2026.12 (2026).
47. M Kwan, R Safavi and Y Wang. Counting perfect matchings in Dirac hypergraphs. *Combinatorica* 46.1 (2026).
46. V Jain, M Kwan, D Mubayi and T Tran. The edge-statistics conjecture for hypergraphs. *International Mathematics Research Notices* 2025.18 (2025).
45. M Kwan and L Sauermann. Resolution of the quadratic Littlewood–Offord problem. *Compositio Mathematica* 161.12 (2026), 3089–3139.
44. M Glasgow, M Kwan, A Sah and M Sawhney. A central limit theorem for the matching number of a sparse random graph. *Journal of the London Mathematical Society* 111.4:e70101 (2025).
43. F Brunck and M Kwan. Books, hallways and social butterflies: a note on sliding block puzzles. *The Mathematical Intelligencer* 47.1 (2025), 52–65.
42. M Anastos, Z Jin, M Kwan, B Sudakov. Extremal, enumerative and probabilistic results on ordered hypergraph matchings. *Forum of Mathematics, Sigma* 13:E55 (2025).
41. M Anastos, O Cooley, M Kang and M Kwan. Partitioning problems via random processes. *Journal of the London Mathematical Society* 110.6:e70010 (2024).
40. M Anastos, B Auerbach, M A Baig, M Cueto Noval, M Kwan, G Pascual-Perez and K Pietrzak. The cost of maintaining keys in dynamic groups with applications to multicast encryption and group messaging. *Proceedings of the Theory of Cryptography Conference (TCC) 2024* (Lecture Notes in Computer Science vol. 15364), 413–443.

39. M Kwan, A Sah, M Sawhney and M Simkin. High-girth Steiner triple systems. *Annals of Mathematics* 200.3 (2024), 1059–1156.
38. M Kwan and Y Wigderson. The inertia bound is far from tight. *Bulletin of the London Mathematical Society*, 56.10 (2024), 3196–3208.
37. I Koval, M Kwan. Exponentially many graphs are determined by their spectrum. *Quarterly Journal of Mathematics* 75.3 (2024), 869–899.
36. M Kwan, A Sah, M Sawhney and M Simkin. Substructures in Latin squares. *Israel Journal of Mathematics* 256.2 (2023), 363–416.
35. M Kwan, A Sah, L Sauermann and M Sawhney. Anticoncentration in Ramsey graphs and a proof of the Erdős–McKay conjecture. *Forum of Mathematics, Pi* 11:E21 (2023).
34. A Ferber, M Kwan, A Sah and M Sawhney. Singularity of the k-core of a random graph. *Duke Mathematical Journal* 172.7 (2023), 1293–1332.
33. J Fox, M Kwan and H Spink. Geometric and o-minimal Littlewood–Offord problems. *Annals of Probability* 51.1 (2023), 101–126.
32. M Kwan, A Sah and M Sawhney. Enumerating matroids and linear spaces. *Comptes Rendus Mathématique* 361 (2023), 565–575.
31. A Ferber, M Kwan, B Narayanan, A Sah and M Sawhney. Friendly bisections of random graphs. *Communications of the American Mathematical Society* 2 (2022), 380–416.
30. A Ferber and M Kwan. Dirac-type theorems in random hypergraphs. *Journal of Combinatorial Theory, Series B* 155 (2022), 318–357.
29. M Kwan, A Sah and M Sawhney. Large deviations in random Latin squares. *Bulletin of the London Mathematical Society* 54.4 (2022), 1420–1438.
28. M Kwan, L Sauermann and Y Zhao. Extension complexity of low-dimensional polytopes. *Transactions of the American Mathematical Society*, 375.6 (2022), 4209–4250.
27. A Ferber, M Kwan and L Sauermann. List-decodability with large radius for Reed–Solomon codes. *IEEE Transactions on Information Theory* 68.6 (2022), 3823–3828. A conference version appeared at *FOCS 2021*.
26. M Kwan and L Sauermann. On the permanent of a random symmetric matrix. *Selecta Mathematica* 8.15 (2022).
25. A Ferber, M Kwan and L Sauermann. Singularity of sparse random matrices: simple proofs. *Combinatorics, Probability and Computing* 31.1 (2022), 21–28.
24. J Fox, M Kwan and L Sauermann. Combinatorial anti-concentration inequalities, with applications. *Mathematical Proceedings of the Cambridge Philosophical Society* 171.2 (2021), 227–248.
23. Z Chroman, M Kwan and M Singhal. Lower bounds for superpatterns and universal sequences. *Journal of Combinatorial Theory, Series A* 156:105467 (2021).
22. J Fox, M Kwan and L Sauermann. Anticoncentration for subgraph counts in random graphs. *Annals of Probability* 49.3 (2021), 1515–1553.
21. J Fox, M Kwan and B Sudakov. Acyclic subgraphs of tournaments with high chromatic number. *Bulletin of the London Mathematical Society* 53.2 (2021), 619–630.
20. A Ferber and M Kwan. Almost all Steiner triple systems are almost resolvable. *Forum of Mathematics, Sigma* 8:E39 (2020).
19. M Bucic, M Kwan, A Pokrovskiy and B Sudakov. Halfway to Rota's basis conjecture. *International Mathematics Research Notices* 2020.21 (2020), 8007–8026.
18. X He and M Kwan. Universality of random permutations. *Bulletin of the London Mathematical Society* 52.3 (2020), 515–529.
17. M Bucic, M Kwan, A Pokrovskiy, B Sudakov, T Tran and A Z Wagner. Nearly-linear monotone paths in edge-ordered graphs. *Israel Journal of Mathematics* 238.2 (2020), 663–685.
16. M Kwan and L Sauermann. An algebraic inverse theorem for the quadratic Littlewood–Offord problem, and an application to Ramsey graphs. *Discrete Analysis* 2020:12 (2020).

15. M Kwan. Almost all Steiner triple systems have perfect matchings. *Proceedings of the London Mathematical Society* 121.6 (2020), 1468–1495.
14. M Kwan, S Letzter, B Sudakov and T Tran. Dense induced bipartite subgraphs in triangle-free graphs. *Combinatorica* 40.2 (2020), 283–305.
13. M Kwan and B Sudakov. Ramsey graphs induce subgraphs of quadratically many sizes. *International Mathematics Research Notices* 2020.6 (2020), 1621–1638.
12. D Conlon, J Fox, M Kwan and B Sudakov. Hypergraph cuts above the average. *Israel Journal of Mathematics* 233.1 (2019), 67–111.
11. M Kwan and B Sudakov. Proof of a conjecture on induced subgraphs of Ramsey graphs. *Transactions of the American Mathematical Society* 372.8 (2019), 5571–5594.
10. M Kwan, B Sudakov and T Tran. Anticoncentration for subgraph statistics. *Journal of the London Mathematical Society* 99.3 (2019), 757–777.
9. M Krivelevich, M Kwan, P-S Loh and B Sudakov. The random k-matching-free process. *Random Structures and Algorithms* 53.4 (2018), 692–716.
8. A Ferber, M Kwan and B Sudakov. Counting Hamilton cycles in sparse random directed graphs. *Random Structures and Algorithms* 53.4 (2018), 592–603.
7. M Kwan, B Sudakov and P Vieira. Non-trivially intersecting multi-part families. *Journal of Combinatorial Theory, Series A* 156 (2018), 44–60.
6. M Kwan and B Sudakov. Intercalates and discrepancy in random Latin squares. *Random Structures and Algorithms* 52.2 (2018), 181–196.
5. A S Bandeira, A Ferber and M Kwan. Resilience for the Littlewood–Offord Problem. *Advances in Mathematics* 319 (2017), 292–312.
4. C Greenhill, M Isaev, M Kwan and B McKay. The average number of spanning trees in sparse graphs with given degrees. *European Journal of Combinatorics* 63 (2017), 6–25.
3. M Krivelevich, M Kwan and B Sudakov. Bounded-degree spanning trees in randomly perturbed graphs. *SIAM Journal on Discrete Mathematics* 31.1 (2017), 155–171.
2. M Krivelevich, M Kwan and B Sudakov. Cycles and matchings in randomly perturbed digraphs and hypergraphs. *Combinatorics, Probability and Computing* 25.6 (2016), 909–927.
1. C Greenhill, M Kwan and D Wind. On the number of spanning trees in random regular graphs. *Electronic Journal of Combinatorics* 21(1):P1.45 (2014).